

Sahil Hossen

☎ +91-9362125184 | ✉ sahilhossen51@gmail.com | 🌐 sahil-hossen | 📺 Sahil-hossen

SUMMARY

Software Development Engineer (CSE, NIT Agartala) experienced in building production **GenAI systems** (RAG pipelines, LLM apps) and scalable **full-stack and backend services**. Skilled in **Python, TypeScript, React, NestJS, Node.js**, REST APIs, SQL, and **AWS** (Bedrock, Lambda, S3, DynamoDB), with strong testing and CI/CD practices. Comfortable owning features end-to-end — from API design and implementation to automated testing, deployment, and monitoring in production.

EDUCATION

National Institute of Technology, Agartala

Nov 2022 – May 2026

B.Tech, Computer Science and Engineering; CGPA: 7.55

Agartala, India

Sigma School of Excellence, Rajasthan

April 2020 – March 2021

Senior Secondary (XII); Percentage: 87.4%

Rajasthan, India

Jawahar Navodaya Vidyalaya, North Tripura

April 2018 – March 2019

Secondary (X); Percentage: 94.4%

Tripura, India

EXPERIENCE

Amdocs

Jan 2026 – June 2026

Software Development Engineer Intern

Pune, India

- Built **GitSense** (NestJS, React, TypeScript), a **RAG pipeline** with AST code chunking, Titan v2 embeddings, and hybrid kNN/BM25 retrieval over **S3 Vectors** for accurate, source-cited code answers.
- Developed a production **voucher microservice** (NestJS, AWS DynamoDB, Lambda, S3) supporting batch creation, presigned-URL exports, and idempotent redemptions with async wallet settlement.
- Contributed to **Docu AI**, a GenAI tool that auto-generates documentation from PR diffs via **AWS Bedrock** and syncs to Confluence; set up **GitHub Actions CI/CD** and **CloudWatch** monitoring.
- Built a **YAML-based regression test harness** for the RAG pipeline, validating file paths, code snippets, and tool-call outputs on every change.
- Developed the **React** UI for querying the codebase with cited answers; contributed to API design, code reviews, and sprint releases.

National Informatics Centre, Govt of India

June 2025 – July 2025

Software Development Engineer Intern

Agartala, India

- Designed and **tested RESTful Spring Boot APIs** for the RTI registration system; wrote **PostgreSQL** queries and validated correctness across multiple relational tables.
- Performed systematic **API testing** with **Postman** — validated request/response schemas, status codes, and edge cases to ensure data integrity.

PROJECTS

LiveLocator | Node.js, Express, Socket.IO, Leaflet.js, EJS

- Built a real-time location-tracking web app with **Node.js, Express, and Socket.IO** for bidirectional updates; integrated **Leaflet.js** with OpenStreetMap for live user markers.
- Handled edge cases (connection drops, stale sessions, geolocation denials) and tested WebSocket events for multi-user concurrency and graceful reconnects.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C/C++, HTML/CSS

GenAI & LLMs: RAG Pipelines, AI Agents, Embeddings, Prompt Engineering, AWS Bedrock

Frameworks & Backend: React.js, NestJS, Node.js, Express.js, Spring Boot, REST APIs, Microservices

Cloud & DevOps: AWS (S3, Lambda, Bedrock, DynamoDB, CloudWatch), Git, GitHub Actions, CI/CD, PostgreSQL

Testing & Fundamentals: Postman, REST API Validation, Automated Testing, DSA, DBMS, OS, OOP

ACHIEVEMENTS

- JEE Main 2022** — Ranked among **top 1.6 percentile** out of 1.2 million candidates.
- CodeChef** — Rank **329** in Starters 196 among 29,657 participants; 2★ rated (1415).
- Smart India Hackathon** — Team of 5 selected among **top 12 college teams** for project idea and presentation.